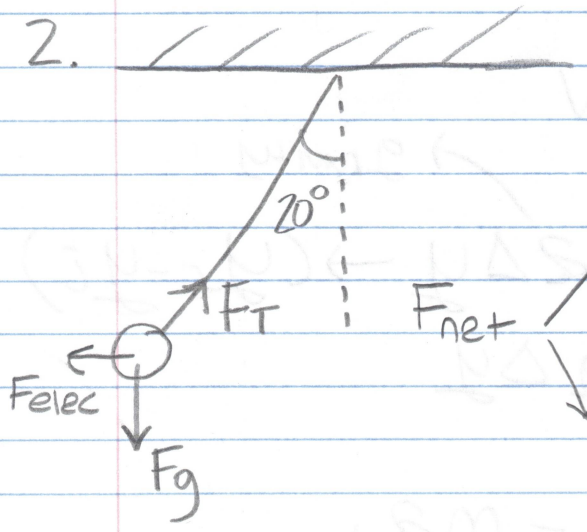
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# PHYS 2500 Lecture 15: "Worksheet 5"

2.




$F_g = -mgy$

$F_{net,x} = 0 = -F_{elec} + F_{Tx}$

$\rightarrow F_{elec} = F_{Tx}$

$F_{net,y} = 0 = -mg + F_{Ty}$

$\rightarrow F_{Ty} = mg$

$\sin\theta = \frac{F_{Tx}}{F_T}$

$F_{Tx} = F_T \sin\theta = \frac{F_{Ty}}{\cos\theta}\sin\theta$

$= F_{Ty}\tan\theta$

$\cos\theta = \frac{F_{Ty}}{F_T}$

$F_T = \frac{F_{Ty}}{\cos\theta}$

$F_{elec} = mg\tan\theta = (9.8\times10^{-3}\,N)(\tan 20^\circ)$

$\boxed{\approx 3.6\times10^{-3}\,N}$

3. Known: $m = 50g$ Final Velocity is $\emptyset$

$h = 1.0m$

$V_i = 2.0N$

gravity

$0 = v_i^2 + 2a\Delta y \rightarrow (y_f - y_i)$

$0 = V_i^2 + 2g\Delta y$

$F_{net,y} = 2.0N = ma_y$

$F_{net} = ma_y$

$$a_y = \frac{F_{net,y}}{m}$$

$V_1^2 = 0 + 2a_y \Delta y_{tube}$

$= 2a_y \Delta y_{tube}$

$$\Delta y = \frac{V_1^2}{2g}$$

$$\Delta y = \frac{(2ay\Delta y + \text{tube})^2}{2g}$$

4. $m = 1.0\,g$

$F_p = 12\,N$

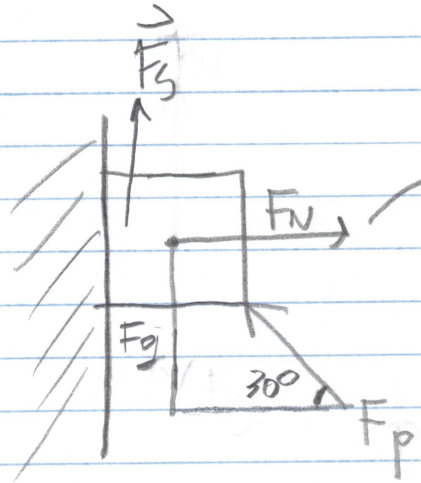
$\theta = 30°$

$\vec{F} = \mu_K n$

$\vec{F}_{s\,max} = \mu_s n$

What is the max static friction applied to the block?

Friction is parallel to surface




→ perpendicular to the surface

$\mu_s$ is assumed to be known for this problem.

$\sin\theta = \frac{F_{py}}{F_p} =$

$\cos\theta = \frac{F_{px}}{F_p} =$

$F_{py} = F_p \sin\theta$

$F_{px} = F_p \cos\theta$

is $F_{py}$ bigger than the force of gravity?